- *LoRaWAN path*—battery-powered sensors in indoor and outdoor locations connect through a LoRa concentrator to a gateway running LoRa Basics Station. This path prioritizes range and power efficiency over data rate, making it ideal for environmental sensors that transmit small amounts of data infrequently. The gateway handles protocol translation, converting LoRa radio signals to internet protocols.
- *WiFi path*—indoor sensors with access to mains power connect directly through the existing WiFi router. This path offers higher data rates and more frequent transmission, suitable for sensors that need to provide richer data streams or more responsive updates.
- *Manual and file integration paths*—both manual readings and CSV file uploads come through a web application that provides a user interface for data entry and file processing. This application serves as a protocol translator, converting human input and file formats into data that can be published to the data collection hub.

The key is that all paths converge on the same transport protocol, MQTT, regardless of how the data was originally collected or transmitted.

3.7.3 The transport layer: unified data flow

The transport layer creates a unified data stream from diverse sources using MQTT as the common protocol. This design provides several important advantages:

- *Protocol abstraction*—the message decoder receives all data through MQTT subscriptions, regardless of whether it originated from a LoRaWAN sensor, WiFi device, manual entry, or file upload and performs any message decoding required. This abstraction simplifies application logic and enables consistent data processing.
- *Scalability*—adding new sensor types or communication methods requires publishing data to MQTT topics and implementing any message decoding that is required. The core application logic remains unchanged.
- *Reliability*—MQTT's publish/subscribe model provides built-in features for handling connection failures, message persistence, and quality of service guarantees that ensure data reaches the application even when network conditions are poor.